```
Atlas atlas-11okly-shard-0 [primary] test> db.students.insertMany([
...   { rollNo: 101, name: "Alice", dept: "CSE", marks: 85, city: "Delhi" },
...   { rollNo: 102, name: "Bob", dept: "ECE", marks: 90, city: "Mumbai" },
...   { rollNo: 103, name: "Charlie", dept: "CSE", marks: 75, city: "Bangalore" },
...   { rollNo: 104, name: "David", dept: "ISE", marks: 60, city: "Chennai" },
...   { rollNo: 105, name: "Eve", dept: "CSE", marks: 45, city: "Hyderabad" }
... ]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69928ee93d9650d21b1e2626'),
    '1': ObjectId('69928ee93d9650d21b1e2627'),
    '2': ObjectId('69928ee93d9650d21b1e2628'),
    '3': ObjectId('69928ee93d9650d21b1e2629'),
    '4': ObjectId('69928ee93d9650d21b1e262a')
  }
}
```

```
Atlas atlas-11okly-shard-0 [primary] test> db.students.find();
[
  {
    _id: ObjectId('69928ee93d9650d21b1e2626'),
    rollNo: 101,
    name: 'Alice',
    dept: 'CSE',
    marks: 85,
    city: 'Delhi'
  },
  {
    _id: ObjectId('69928ee93d9650d21b1e2627'),
    rollNo: 102,
    name: 'Bob',
    dept: 'ECE',
    marks: 90,
    city: 'Mumbai'
  },
  {
    _id: ObjectId('69928ee93d9650d21b1e2628'),
    rollNo: 103,
    name: 'Charlie',
    dept: 'CSE',
    marks: 75,
    city: 'Bangalore'
  },
  {
    _id: ObjectId('69928ee93d9650d21b1e2629'),
    rollNo: 104,
    name: 'David',
    dept: 'ISE',
    marks: 60,
    city: 'Chennai'
  },
  {
    _id: ObjectId('69928ee93d9650d21b1e262a'),
    rollNo: 105,
    name: 'Eve',
    dept: 'CSE',
    marks: 45,
    city: 'Hyderabad'
  }
]
```

```
Atlas atlas-11okly-shard-0 [primary] test> db.students.updateOne(
...   { rollNo: 101 },
...   { $set: { marks: 95 } }
... );
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Atlas atlas-11okly-shard-0 [primary] test> db.students.deleteOne({ rollNo: 105 });
{ acknowledged: true, deletedCount: 1 }
Atlas atlas-11okly-shard-0 [primary] test> db.students.find();
[
  {
    _id: ObjectId('69928ee93d9650d21b1e2626'),
    rollNo: 101,
    name: 'Alice',
    dept: 'CSE',
    marks: 95,
    city: 'Delhi'
  },
  {
    _id: ObjectId('69928ee93d9650d21b1e2627'),
    rollNo: 102,
    name: 'Bob',
    dept: 'ECE',
    marks: 90,
    city: 'Mumbai'
  },
  {
    _id: ObjectId('69928ee93d9650d21b1e2628'),
    rollNo: 103,
    name: 'Charlie',
    dept: 'CSE',
    marks: 75,
    city: 'Bangalore'
  },
  {
    _id: ObjectId('69928ee93d9650d21b1e2629'),
    rollNo: 104,
    name: 'David',
    dept: 'ISE',
    marks: 60,
    city: 'Chennai'
  }
]
```

```
Atlas atlas-11okly-shard-0 [primary] test> db.students.deleteMany({ marks: { $lt: 40 } });
{ acknowledged: true, deletedCount: 0 }
Atlas atlas-11okly-shard-0 [primary] test> db.students.find({
...   dept: "CSE",
...   marks: { $gt: 80 }
... });
[
  {
    _id: ObjectId('69928ee93d9650d21b1e2626'),
    rollNo: 101,
    name: 'Alice',
    dept: 'CSE',
    marks: 95,
    city: 'Delhi'
  }
]
Atlas atlas-11okly-shard-0 [primary] test> db.students.find({
...   dept: { $in: ["CSE", "ECE"] }
... });
[
  {
    _id: ObjectId('69928ee93d9650d21b1e2626'),
    rollNo: 101,
    name: 'Alice',
    dept: 'CSE',
    marks: 95,
    city: 'Delhi'
  },
  {
    _id: ObjectId('69928ee93d9650d21b1e2627'),
    rollNo: 102,
    name: 'Bob',
    dept: 'ECE',
    marks: 90,
    city: 'Mumbai'
  },
  {
    _id: ObjectId('69928ee93d9650d21b1e2628'),
    rollNo: 103,
    name: 'Charlie',
    dept: 'CSE',
    marks: 75,
    city: 'Bangalore'
  }
]
```

```
Atlas atlas-11okly-shard-0 [primary] test> db.students.find(
...   { marks: { $gte: 60, $lte: 90 } },
...   { name: 1, marks: 1, _id: 0 }
... );
[
  { name: 'Bob', marks: 90 },
  { name: 'Charlie', marks: 75 },
  { name: 'David', marks: 60 }
]
Atlas atlas-11okly-shard-0 [primary] test> db.students.insertOne({
...   rollNo: 106, name: "Frank", dept: "CSE", marks: 70, city: null
... });
{
  acknowledged: true,
  insertedId: ObjectId('699291173d9650d21b1e262b')
}
Atlas atlas-11okly-shard-0 [primary] test> db.students.find({ city: null });
[
  {
    _id: ObjectId('699291173d9650d21b1e262b'),
    rollNo: 106,
    name: 'Frank',
    dept: 'CSE',
    marks: 70,
    city: null
  }
]
Atlas atlas-11okly-shard-0 [primary] test> db.students.find({ city: { $exists: false } });

Atlas atlas-11okly-shard-0 [primary] test> db.students.updateMany(
...   { city: null },
...   { $set: { city: "Unknown" } }
... );
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

```
Atlas atlas-11okly-shard-0 [primary] test> db.students.countDocuments();
5
Atlas atlas-11okly-shard-0 [primary] test> db.students.countDocuments({ dept: "ISE" });
1
Atlas atlas-11okly-shard-0 [primary] test> db.students.countDocuments({ marks: { $gt: 75 } });
2
Atlas atlas-11okly-shard-0 [primary] test> db.students.find().sort({ marks: -1 });
[
  {
    _id: ObjectId('69928ee93d9650d21b1e2626'),
    rollNo: 101,
    name: 'Alice',
    dept: 'CSE',
    marks: 95,
    city: 'Delhi'
  },
  {
    _id: ObjectId('69928ee93d9650d21b1e2627'),
    rollNo: 102,
    name: 'Bob',
    dept: 'ECE',
    marks: 90,
    city: 'Mumbai'
  },
  {
    _id: ObjectId('69928ee93d9650d21b1e2628'),
    rollNo: 103,
    name: 'Charlie',
    dept: 'CSE',
    marks: 75,
    city: 'Bangalore'
  },
  {
    _id: ObjectId('699291173d9650d21b1e262b'),
    rollNo: 106,
    name: 'Frank',
    dept: 'CSE',
    marks: 70,
    city: 'Unknown'
  },
  {
    _id: ObjectId('69928ee93d9650d21b1e2629'),
    rollNo: 104,
    name: 'David',
    dept: 'ISE',
    marks: 60,
    city: 'Chennai'
  }
]
```

```
Atlas atlas-11okly-shard-0 [primary] test> db.students.find().sort({ marks: -1 }).limit(3);
[
  {
    _id: ObjectId('69928ee93d9650d21b1e2626'),
    rollNo: 101,
    name: 'Alice',
    dept: 'CSE',
    marks: 95,
    city: 'Delhi'
  },
  {
    _id: ObjectId('69928ee93d9650d21b1e2627'),
    rollNo: 102,
    name: 'Bob',
    dept: 'ECE',
    marks: 90,
    city: 'Mumbai'
  },
  {
    _id: ObjectId('69928ee93d9650d21b1e2628'),
    rollNo: 103,
    name: 'Charlie',
    dept: 'CSE',
    marks: 75,
    city: 'Bangalore'
  }
]
```

```
Atlas atlas-11okly-shard-0 [primary] test> db.students.find().skip(2).limit(3);
[
  {
    _id: ObjectId('69928ee93d9650d21b1e2628'),
    rollNo: 103,
    name: 'Charlie',
    dept: 'CSE',
    marks: 75,
    city: 'Bangalore'
  },
  {
    _id: ObjectId('69928ee93d9650d21b1e2629'),
    rollNo: 104,
    name: 'David',
    dept: 'ISE',
    marks: 60,
    city: 'Chennai'
  },
  {
    _id: ObjectId('699291173d9650d21b1e262b'),
    rollNo: 106,
    name: 'Frank',
    dept: 'CSE',
    marks: 70,
    city: 'Unknown'
  }
]
Atlas atlas-11okly-shard-0 [primary] test> db.students.aggregate([
...   { $group: { _id: null, avgMarks: { $avg: "$marks" } } }
... ]);
[ { _id: null, avgMarks: 78 } ]
Atlas atlas-11okly-shard-0 [primary] test> db.students.aggregate([
...   { $group: { _id: "$dept", totalMarks: { $sum: "$marks" } } }
... ]);
[
  { _id: 'ECE', totalMarks: 90 },
  { _id: 'ISE', totalMarks: 60 },
  { _id: 'CSE', totalMarks: 240 }
]
```

```
Atlas atlas-11okly-shard-0 [primary] test> db.students.aggregate([
...   { $group: { _id: "$dept", maxMarks: { $max: "$marks" } } }
... ]);
[
  { _id: 'CSE', maxMarks: 95 },
  { _id: 'ISE', maxMarks: 60 },
  { _id: 'ECE', maxMarks: 90 }
]
```